87

15

Roll No. | | | | | | | |

Set-1

END SEMESTER EXAMINATION 2024

Name of the Course: B.Tech Petroleum Engineering *Semester: VI*

Name of the Paper: Reservoir Modeling & Simulation *Course Code:* TPE-601

Time: Three Hours MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		COs
a.	What is workflow and sequential workflow in context with the modeling process? What are the Essential data required for building a reservoir model?	CO1
b.	What is Surface & Interfacial tension explain with diagram? What is wetting & non wetting phase of a reservoir?	CO1
c.	What is sedimentary rock? How many type of these rocks are available? Write and explain Minimum five important physical properties of the rocks.	CO2
Q2		
a.	What is petroleum system? Explain with diagram showing Generation, Migration and trapping of hydrocarbons of a Petroleum system.	CO2
b.	What are the various type of data required for a reservoir model building ? Explain in Details.	CO2
c.	Explain the various steps of data import, Visualization, data editing & Interpretation Well Correlation of various sequences?	CO3
Q3		
a.	What is the difference between classic reservoir engineering & Reservoir modeling & simulation?	CO1
b.	Categorize and explain in detail the various types of data required for facies log creation, fault modeling, structure & Property model.	CO4
c.	Construct a suitable workflow for fault and structure model building of a reservoir model in RMS.	CO3
Q4		
a.	What are Cubical packing, orthorhombic-packing & rhombohedra –packing of spheres. What are the porosity values in all type of packing?	CO2
b.	Explain porous media. Write Darcy's equation for fluid flow in a porous media.What are its limitations?	CO2
c.	What is facies modeling? Design a workflow for Facies modeling. Explain its all components. Why it is required?	CO4

PTO →

A1

Q5		
a.	What are the different category of reserve? What are the various techniques for reserve Estimation? What are the different stages for reserve estimation?	CO5
b.	What do you understand by upscale? What are the different methods for up-scaling the Petrophysical property Porosity, Saturation & Permeability. Give the example of each.	CO4
c.	What is Petrophysical modeling? Compare deterministic & Stochastic methods for Petrophysical modeling. Where these methods will be recommended?	CO5

Roll No. | | | | | | | |

88

20

EVEN END SEMESTER EXAMINATION 2024

Name of the Program: **B.Tech. (PE)**
Name of the Course: **Data Science**
Time: **3 Hours**

Semester: **6th**
Course Code: **TPE-602**
Maximum Marks: **100**

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub-questions among a, b & and c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(20marks)**	
(a)	Discuss the various problem-solving frameworks and techniques in data science.	CO1
(b)	What are the emerging trends and future challenges in the field of data science, and how do they shape the landscape of data-driven decision-making?	CO1, CO2
(c)	Classify various data science models based on their working capabilities.	CO1
Q2	**(20 marks)**	
(a)	Consider a table named customers with the following columns: customer_id, name, age, and city. • Write an SQL query to retrieve the names of all customers. • Write an SQL query to retrieve the names and ages of customers residing in the city of 'New York'. • Write an SQL query to calculate the average age of all customers.	CO2
(b)	The given dataset 'PE_ENGG_SALES.csv', contains information about sales transactions. The dataset includes the following columns: transaction_id, product_id, customer_id, amount, and date. Your task is to write SQL queries to perform the following tasks: • Retrieve transactions where the amount exceeds $100. • Retrieve transactions made by customers with IDs between 1000 and 2000.	CO2, CO1
(c)	Differentiate between Group By and Having clause with the help of suitable example.	CO2
Q3	**(20 marks)**	
(a)	Explain popular data visualization techniques used in Excel.	CO3
(b)	Discuss about the hypothesis testing? Clearly states the differences between basic and advanced hypothesis testing approaches.	CO1
(c)	Write short notes on: (i) Inferential Statistics (ii) Bayes theorem and its significance	CO3
Q4	**(20 marks)**	
(a)	State the significance of the R language in data science. Also, explain steps used in data processing analysis.	CO4
(b)	Write short notes on: (i) Predictive analysis in R (ii) Data Visualization techniques	CO2, CO4
(c)	Discuss the importance of time series forecasting in various domains such as finance, weather prediction, and supply chain management. Explain the key techniques used for time series analysis and forecasting, highlighting their strengths and limitations	CO3
Q5	**(20 marks)**	
(a)	Why is data manipulation required before fitting a machine learning model? Also, explain feature engineering techniques used in data science.	CO5
(b)	Differentiate and classify the various clustering and classification data science models based on their working capabilities.	CO4
(c)	Describe significant techniques for improving the interpretability in data science models in real-world applications.	CO4, CO5

89

15

Roll No. | | | | | | | |

END SEMESTER EXAMINATION 2024

Name of the Course: B.Tech Petroleum Engineering
Name of the Paper: Natural Gas Engineering

Semester: VI
Course Code: TPE-603

Time: Three Hours

MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1

		COs
a.	Discuss the properties of natural gas? A natural gas has the following compositions.	CO1, CO2

Composition	Mole fractions (yi)	MWi
CO2	0.02	44
CH4	0.93	16
C2H6	0.03	30
C3H8	0.02	44

Assume compressibility factor Z = 0.82
The Universal gas constant = 10.73 psi ft3/lb-mol ^{0}R
Density of natural gas at 2000 psia and 180^{0}F is ------------ lb./ft3 (**rounded off to two decimal places**)

		COs
b.	What are the processes/methods of conventional natural gas production? Write down the steps in well completion. Solve the numerical	CO1, CO2

Composition	Mole fractions (yi)	MWi	μi (cp)
CO2	0.05	44	0.0141
CH4	0.95	16	0.0159

Assuming Ideal gas behavior. Find the gas viscosity by using mixing rule at 320 K.

		COs
c.	What do you mean by clathrates? Describe in details about all the properties of gas hydrate. Draw boundaries with respect to depth and temperature for different phase by stating the different region as methane gas, methane gas hydrates water and ice.	CO3

Q2		
a.	A gas well is producing a natural gas with the following composition: Component / y_i CO_2 0.05 C_1 0.90 C_2 0.03 C_3 0.02 Assuming Ideal gas behavior, Calculate the following: a. Apparent molecular weight of the gas b. Gas specific gravity c. Gas density at 2000psi and 150 degree F d. Specific volume at 2000psi and 150 degree F e. Calculate Gas density at 2000psi and 150 degree F considering real gas by taking Z=0.85	CO3, CO4
b.	Explain hydraulic fracturing? Discuss the hydraulic fracturing process? Discuss in details about the fracturing fluid by defining the base fluid, surfactant and non-emulsifiers. **solve** Which of the following is the primary role of proppants in hydraulic fracturing? a. Keep the fractures open during production b. Decrease the viscosity of fracturing fluid c. Decrease the density of fracturing fluid d. Reduce the viscosity of crude oil in reservoir	CO1, CO4
c.	What are the conventional natural gas production systems? Write down the conditions in terms of pressure with respect to bubble point and dew point for the black oil, volatile oil, solution gas drive, wet gas and gas condensate with the help of phase diagram	CO1, CO2
Q3		
a.	Classify and discuss the Sweetening process? What are the different Amines used in the gas amine based sweetening process? Define iron-sponge sweetening? Write down the reactions involved in the iron sponge process	CO1
b.	What specific information's should be gathered to design the glycol dehydrator? Discuss. What are the two design criteria to be employed for the glycol dehydrator?	CO1, CO2
c.	Why glycol/Amine process is used in sweetening of natural gas? Discuss the proportions of solution being used in glycol/Amine process. What are the main advantage and disadvantage of glycol/Amine process in the sweetening of natural gas	CO2, CO5
Q4		
a.	Draw the flow sheet diagram of Selexol process in the sweetening of natural gas with their process description.	CO1, CO5
b.	Explain the direct recoding charts and square root charts with neat diagram in the measurement of natural gas	CO4, CO5
c.	Illustrate the orifice equations, explain all the factors used in the equation in the measurement of the natural gas	CO3, CO6
Q5		

a.	Draw the transportation system? Pipeline design equation in transportation of natural gas depends on which factor? Write down the complex equations which have been developed for sizing natural gas pipelines in various flow conditions?	CO4, CO6
b.	Calculate the maximum quantity of gas that can be transported through 4" dia, 20 Km long horizontal pipeline under the following conditions : Inlet pressure = 2000 psia Maximum pressure drop = 500 psi Average gas temp. = 100 F Pipeline efficiency factor = 0.92 Gas gravity = 0.6 Zavg can be taken as = 0.9 Use both Panhandle B and Weymouth methods	CO4, CO5
c.	Explain Corrosion in Natural Gas Transportation and their mitigation in: i. external corrosion mitigation ii. internal corrosion mitigation	CO2, CO3

91

Roll No.

END SEMESTER EXAMINATION JUNE- 2024

Name of The Program: ***B.Tech (Petroleum Engineering)*** Semester: **VI**

Name of The Course: **Petroleum Production Engineering-II** Course Code: **TPE-605**

Time: 3 Hours Maximum Marks: **100**

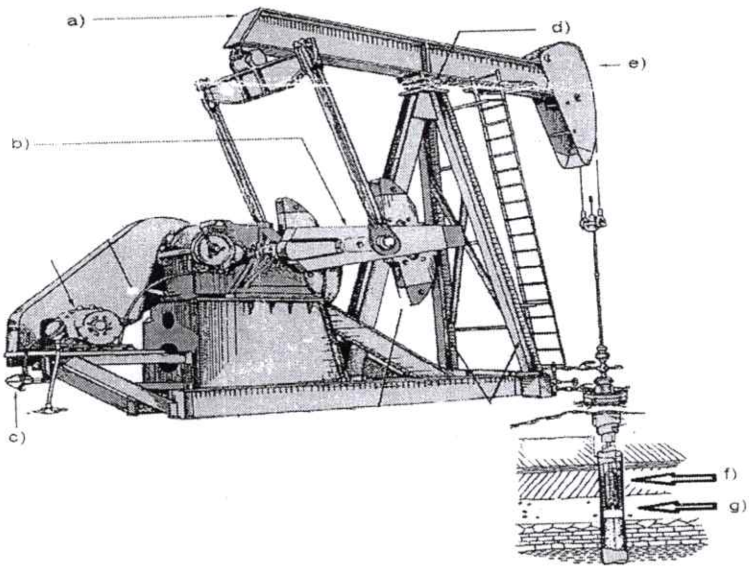
Note:
(i) All Questions are compulsory
(ii) Answer any two questions among a, b & c in each main question.
(iii) Total marks each main question is twenty.
(iv) Each Question carries 10 Marks

Q1. **(20 Marks)(CO1,4)**

a) Explain the load which are temporary in nature/on platform.

b) Describe the platform which rests on 4-6 pillar like legs.

c) From the given figure identify & explain type of Artificial lift and also identify parts (a, b, c, d, e, f, g) in the given figure, also write why it is used as artificial lift.



Q2. **(20 Marks) (CO2,4,5)**

a) Explain what you understand by perforation, types of guns and explosive used in perforation.

b) Define EOR and also describe thermal method of EOR.

c) Explain the concept of reservoir deliverability and illustrate all the several factors on which reservoir deliverability depends?

Q3. **(20 Marks)(CO2,3)**

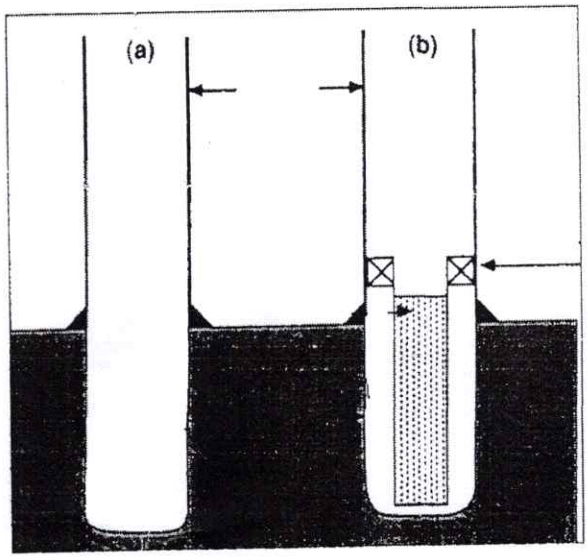
a) Define well activation, explain the concept of swabbing and why swabbing is necessary process in well Activation

b) Explain the following: -
i) Gas Well Deliverability
ii) Choke Performance
iii) Oil Well Performance

c) Illustrates the nodes in an oil and gas production system and pressure trend along the flow path and also write all the abbreviations used for making nodes. and also describe what is nodal analysis and it is used in Petroleum Production.

Q4. **(20 Marks)(CO2,5)**

a) From the given figure identify types of liner and differentiate between them & also identify & name the parts of liner which are shown by arrows.



b) Describe enhance oil recovery and write all the methods off enhance oil recovery and explain any one method.

c) Differentiate Between Gas injection and Water Injection

Q5 **(20 Marks) (CO5,6)**

a) With the help of a diagram explain intermittent gas flow system and Progressive cavity pumps system.

b) Write the advantages and disadvantages of hydraulic jet pumps and gas lift system.

c) Explain the artificial lift method which can be used for volumes range from a low of 150 B/D.

92

Roll No. | | | | | | | |

END SEMESTER EXAMINATION 2024

Name of the Program: B.Tech Petroleum Engineering *Semester: VI*

Name of the Course: Alternate Energy Resources *Course Code:* TPE-611

Time: Three Hours

MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		COs
a.	Discuss in details about the energy sources and their availability with respect to commercial or conventional energy resources	CO1, CO2
b.	Write a short note on the classification of solar energy storage systems	CO2, CO3
c.	What are the renewable energy resources? Discuss advantages and prospects of renewable energy resources	CO3
Q2		
a.	What do you mean by fluidized bed technology for the coal gasification? Discus the principle of fluidization and fluidized bed combustion	CO3, CO4
b.	Illustrate the principle of solar pond with the help of neat and labeled diagram?	CO1, CO4
c.	Illustrate the following in solar photovoltaic (SPV) systems i. Semiconductors ii. Intrinsic and extrinsic semiconductors iii. N-Type, P-Type semiconductor iv. Photovoltaic effects	CO1, CO2, CO4
Q3		
a.	What are the advantages and disadvantages of hydro-electric power plant? Discuss in detail about the selection of site for hydro-electric power plant?	CO1, CO3
b.	Illustrate the classification of hydro-electric power plants with neat and labelled diagram?	CO1, CO3
c.	Explain hydraulic turbines? Discuss classification of hydraulic turbines? Explain the calculation of available hydro-power with their mathematical expression?	CO2, CO5
Q4		
a.	Discuss principle of wind energy conversion and wind power? Define the following:	CO1, CO5

	i. Gilberts limit ii. Wind energy pattern factor (EPF) iii. Performance of wind mills	
b.	Explain in details about the basic components of wind energy conversion factor. Discus the different terms and definition frequently used for wind energy	CO4, CO5
c.	The following data relates to a propeller turbine: Velocity of wind at 20ºC = 20m/s Turbine diameter 12 m Operating speed of the turbine = 45r.p.m at maximum efficiency Calculate: i. Total power density in the wind stream ii. Maximum obtainable power density iii. Reasonably obtainable power density iv. Total power generated v. Maximum torque and maximum axial thrust.	CO3, CO6
Q5		
a.	What is biomass? What are the different biomass resources? Discuss the different biomass conversion processes?	CO1 CO4,
b.	Define biogas plants? Discuss the raw materials used and their main components of a biogas plant	CO4, CO5
c.	Explain production of biofuels as: i. Production of ethanol ii. Production of methanol	CO2, CO3